Course of AI Batch 2

Week 6 · Assignment

Unmesh

October 21, 2026

This document contains the questions for the Week 6 assignment.

Question 1

Two friends play a match of chess games. The first to win 5 games takes the whole pot of 96 coins. Each game has a winner (no draws) and both players are equally strong, so each game is a fair coin toss. The match is stopped when Asha has won 3 games and Bhanu has won 1.

(a) Following Fermat's idea from Lecture 16, find the largest number of further games that could be needed to settle the match. Write the sample space Ω of these games as a Cartesian product and give $|\Omega|$. Then argue that Ω has exactly as many elements as the power set of $\{1, 2, 3, 4, 5\}$, and say which subsets correspond to Asha winning the match.

(b) Count the outcomes in which Asha wins the match and find how the 96 coins should be split. Pascal objected that Fermat "plays games that would never be played". Settle this by computing Asha's chance a second time, where the match stops the moment someone reaches 5 wins. Explain why the two answers agree.

(c) Bhanu's friend says, "Just split the pot in the ratio of games won, 3:1." Argue whether this is fair. Give at least two concrete situations where the ratio rule clearly gives a wrong or meaningless answer.

(d) Now suppose Asha is the stronger player and wins each game with probability 2/3. Explain why the formula $P(E) = |E|/|\Omega|$ from part (b) can no longer be used, and find Asha's correct share of the pot as a probability.

Question 2

An online store looked at 400 orders from last month. Let D be the event that a discount coupon was used and R the event that the order was returned.

	Returned (R)	Kept (R')	Total
Coupon (D)	48	112	160
No coupon (D')	24	216	240
Total	72	328	400

(a) Draw the Venn diagram with the count in each region. Find $P(D \cap R)$, $P(D \setminus R)$, $P(R \setminus D)$ and $P(D \cup R)$. Check your answer for $P(D \cup R)$ using $P(D) + P(R) - P(D \cap R)$ and explain why the last term must be subtracted. Verify De Morgan's law $(D \cup R)^c = D^c \cap R^c$ using the counts.

(b) Find $P(R \mid D)$ and $P(D \mid R)$. The manager reads the report and says, "Two thirds of our returns came with a coupon, so coupons make two thirds of buyers return their orders." Argue and explain what is wrong with this statement. In your answer, say which sample space each of the two conditional probabilities lives in.

(c) Next month the store expects 70% of orders to use a coupon. Assume the return rate among coupon orders and among non-coupon orders stays the same. Use the law of total probability to predict the overall return rate. Also find the new $P(D \mid R)$. It has changed although nothing about coupon buyers changed. Explain why.

(d) A student claims that $P(R \mid D) + P(R \mid D') = 1$ because "D and D' cover everything". Another claims that $P(R \cap D)$ can be larger than $P(R \mid D)$. Argue whether each claim is true in general, and support your argument with the numbers in the table.

Question 3

A bank's system checks card transactions for fraud. From past data, 1 transaction in 20 is fraud, so the prior is $P(F) = 0.05$. Let F' mean the transaction is genuine. The system looks at three pieces of evidence E_1, E_2, E_3. The table gives the chance of seeing each one when the transaction is fraud and when it is genuine.

evidence E_i	$P(E_i \mid F)$	$P(E_i \mid F')$
E_1: made from a new device	0.6	0.1
E_2: amount unusually large	0.5	0.05
E_3: made in the customer's usual city	0.2	0.8

A transaction arrives and all three are observed, in the order E_1, E_2, E_3.

(a) Update the belief after E_1 using Bayes' theorem in full, finding the evidence term $P(E_1)$ with the law of total probability. An analyst says, "The system says 0.6 for a new device, so there is a 60% chance this is fraud." Argue and explain the mistake.

(b) Carry out all three updates using odds and likelihood ratios. Give the prior odds, each LR, the odds and the probability after each step. Then check your final answer with a natural frequency tree that starts with 20,000 transactions. For each piece of evidence, say whether it raised or lowered the belief and why.

(c) Show that the final belief is the same if the evidence arrives in the order E_3, E_2, E_1. What assumption about E_1, E_2, E_3 have we made by multiplying the likelihood ratios one after another? Argue whether this assumption is reasonable here. If fraudsters who use a new device also tend to make large purchases, will the method overestimate or underestimate the fraud probability? Explain.